

COSC 594/650

Convergence Analysis

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Iterative descent template

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Direction: d_k

Stepsize: $\alpha_k > 0$

An algorithm is a direction rule plus a step-size rule

■ CONVERGENCE METRIC

$$e(x_k) = \|x_k - x^\star\| \quad \text{or} \quad e(x_k) = f(x_k) - f(x^\star)$$

Global convergence to stationary points

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rules: constant, diminishing, Armijo backtracking

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x} \implies \bar{x} \text{ is a stationary point}$

Every limit point $\bar{x}$ is stationary: $\nabla f(\bar{x}) = 0$

Convexity is not required!

Why the stationarity theorem works?

■ DESCENT LEMMA

$$f(x+h) - f(x) \leq h^\top \nabla f(x) + \frac{L}{2} \|h\|^2, \quad \forall h \in \mathbb{R}^n$$

★ $f(x_k + \alpha_k d_k) - f(x_k) \leq \alpha_k \nabla f(x_k)^\top d_k + \frac{(\alpha_k)^2 L}{2} \|d_k\|^2$ – one-step improvement

One-step improvement for steepest descent:

$$f(x_k + \alpha_k d_k) - f(x_k) \leq -\frac{1}{2L} \|\nabla f(x_k)\|^2, \quad \alpha_k = \frac{1}{L}$$

Stationarity theorem: Diminishing step-size

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rule: $\alpha_k \rightarrow 0, \sum_k \alpha_k = \infty$

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x} \implies \bar{x} \text{ is a stationary point}$

PROOF: By contradiction

Stationarity theorem: Armijo backtracking

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rule: Armijo backtracking

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x} \implies \bar{x} \text{ is a stationary point}$

PROOF: By contradiction

Stationarity theorem: Constant step-size

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rule: $0 < \epsilon \leq \alpha_k \leq \frac{(2 - \epsilon)|\nabla f(x_k)^\top d_k|}{L\|d_k\|^2}$

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x} \implies \bar{x} \text{ is a stationary point}$

PROOF: By contradiction

Quadratic example: exact dynamics

Quadratic function: $f(x) = \frac{1}{2}x^\top Qx$, $Q = Q^\top \succ 0$

Steepest descent: $x_{k+1} = (I - \alpha_k Q)x_k$, $k = 0, 1, 2, \dots$

■ OPTIMAL SOLUTION

$$\nabla f(x^\star) = 0 \quad \Longleftrightarrow \quad x^\star = 0$$

■ ERROR DYNAMICS

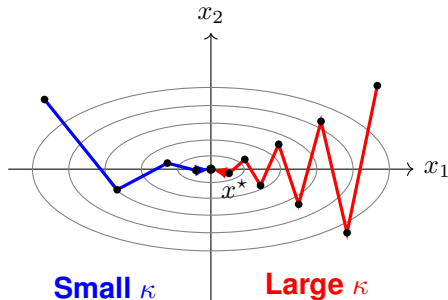
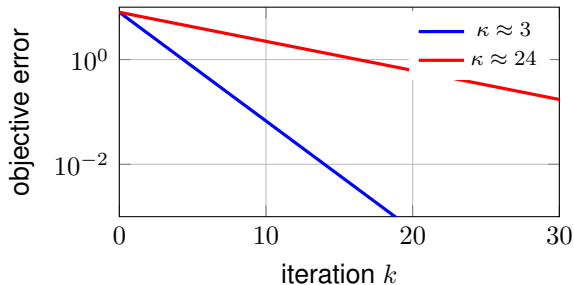
$$\frac{\|x_{k+1}\|^2}{\|x_k\|^2} \leq \|I - \alpha_k Q\|^2 = \max \{ |1 - \alpha_k \lambda_{\min}(Q)|^2, |1 - \alpha_k \lambda_{\max}(Q)|^2 \}$$

★ $\alpha_k = \frac{2}{\lambda_{\min}(Q) + \lambda_{\max}(Q)}$ – best constant step-size

$$\text{Linear rate: } \frac{\|x_{k+1} - x^\star\|^2}{\|x_k - x^\star\|^2} \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^2, \quad \kappa = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}$$

Conditioning controls speed and trajectory

$$e(x_k) \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^{2k} e(x_0), \quad e(x_k) = f(x_k) - f(x^*)$$



$$\text{Iteration complexity: } \mathcal{O} \left(\kappa \log \frac{e(x_0)}{\epsilon} \right)$$

Convexity vs strong convexity

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (differentiable)

■ CONVEX FUNCTION

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

★ $\nabla^2 f(x) \succeq 0$ – PSD

■ STRONGLY CONVEX FUNCTION

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2$$

★ $\nabla^2 f(x) \succeq \mu I$ – PD

★ $\mu I \preceq \nabla^2 f(x) \preceq LI$ – PD and Lipschitz gradient (L)

Questions?

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